

Queue Layouts on Generalized Petersen Graphs $(2k + 1, k)^*$

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Abstract

A queue layout of a graph consists of a linear order of its vertices and a partition of its edges into queues, such that no two edges in the same queue are nested. The minimum number of queues in a queue layout of a graph G , denoted by $qn(G)$, is called the queue number of G . In this paper, we deal with the queue layout of generalized Petersen graphs, denoted by $GP(n, k)$, and contribute some results as follows:

- (1) $qn(GP(3, 1)) = 1$.
- (2) $qn(GP(2k + 1, k)) = 2$ if $k \geq 2$.
- (3) $qn(GP(2k + 1, 2)) = 2$ if $k \geq 2$.

Keywords: Combinatorial problems; Queue layout; Generalized Petersen graph; Interconnection networks;

1. Introduction

Let G be a graph with vertex set $V(G)$ and edge set $E(G)$. A vertex ordering σ of G is a bijection from $V(G)$ to $\{1, 2, \dots, |V(G)|\}$. For $u, v \in V(G)$, we write $u <_\sigma v$ if $\sigma(u) < \sigma(v)$. An m -queue layout of a graph G consists of a vertex ordering σ and a partition of its edges into m queues such that no two edges in the same queue are nested (i.e., two edges $(u, v), (x, y) \in E(G)$ are nested if $u <_\sigma x <_\sigma y <_\sigma v$).

$y <_\sigma v$). The queue number of a graph G , denoted by $qn(G)$, is the minimum m such that G has an m -queue layout. A graph G is an m -queue graph if $qn(G) \leq m$.

Queue layouts were first introduced by Heath et al. [6, 9]. There are many applications of queue layout in computer science, including sorting permutations, parallel process scheduling, matrix computations, and graph drawing (see [1, 9, 14] and references therein quoted). In particular, queue layouts of interconnection networks have applications to the Diogenes approach to testable fault-tolerant arrays of processors [14]. Heath and Rosenberg [9] showed that the problem of recognizing m -queue graphs is NP-complete even if $m = 1$. Thus, further investigations tended to study bounds on the queue number for certain families of graphs [1–13, 16, 17].

The Petersen graph is certainly one of the most famous objects that graph theorists have come across. In [15], Watkins introduced the notion of generalized Petersen graphs, denoted by $GP(n, k)$. Assume that n and k are positive integers with $n \geq 2k + 1$. The $GP(n, k)$ is the graph with vertex set $A \cup B$, where $A = \{a_i \mid 0 \leq i \leq n - 1\}$ and $B = \{b_i \mid 0 \leq i \leq n - 1\}$, and edge set $E_1 \cup E_2 \cup E_3$, where $E_1 = \{a_i a_{i+1} \mid 0 \leq i \leq n - 1\}$, $E_2 = \{a_i b_i \mid 0 \leq i \leq n - 1\}$, and $E_3 = \{b_i b_{i+k} \mid 0 \leq i \leq n - 1\}$. Here the subscripts are to be read as integers modulo n . For example, Figure 1 depicts $GP(3, 1)$ and $GP(5, 2)$. To the best of our knowledge, this article is the first paper to discuss the queue number of generalized Petersen graphs.

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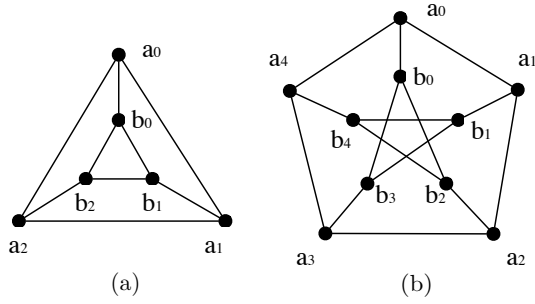


Figure 1: (a) $GP(3, 1)$; (b) $GP(5, 2)$.

2 Queue layout of $GP(2k + 1, k)$

We describe some definitions and introduce some additional notations. Let K_n denote the complete graph with n vertices, and let $CI_n(j_1, j_2)$ denote the circulant graph with n vertices in which the i th vertex is adjacent to the $(i+j_1)$ th, $(i-j_1)$ th, $(i+j_2)$ th, and $(i-j_2)$ th vertices. An *edge contraction* is an operation which removes an edge from a graph while simultaneously merging together the two vertices it previously connected. Without loss of generality, the resulting vertex of contracting edge (x, y) is x . A graph H is a *minor* of a graph G if a copy of H can be obtained from G via repeated edge deletion and/or edge contraction.

Given a vertex ordering σ of a graph $G = (V, E)$, the *length* of an edge $(u, v) \in E$ is defined to be $\ell_\sigma(u, v) = |\sigma(u) - \sigma(v)|$. Note that if $|\ell_\sigma(u, v) - \ell_\sigma(x, y)| \leq 1$, then (u, v) and (x, y) do not nest.

Lemma 1 $qn(GP(3, 1)) = 1$.

Proof. As shown in Figure 1(a), $GP(3, 1)$ contains 6 vertices and 9 edges. To prove $qn(GP(3, 1)) = 1$, we have found a vertex ordering and show that all 9 edges can be put into one queue such that no two edges are nested. The desired vertex ordering is $a_0, a_1, b_0, a_2, b_1, b_2$ and Figure 2 shows that all 9 edges can be put into the same queue. It is not hard to verify the validity of the drawing. $\square$

Lemma 2 $qn(GP(2k + 1, k)) \leq 2$ if $k \geq 2$.

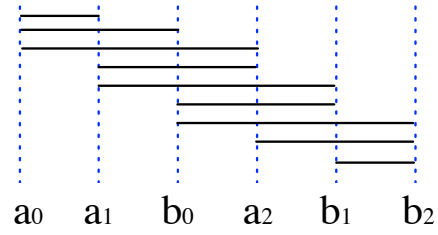


Figure 2: A 1-queue layout of $GP(3, 1)$.

Proof. We now show that the vertex ordering $a_0, a_1, a_2, \dots, a_{2k}, b_0, b_1, b_2, \dots, b_{2k}$ permits a 2-queue layout. All edges in $E_1 - (a_0, a_{2k}) \cup E_3$ are assigned to the first queue and all edges in $E_2 \cup (a_0, a_{2k})$ are assigned to the second queue.

Firstly, since all edges in $E_1 - (a_0, a_{2k})$ have the same length one, any two of them do not nest. The length of the edge in E_3 is either k or $k + 1$, so any two of them do not nest. Due to $a_i <_\sigma b_j$ for $1 \leq i, j \leq n$, any edge in $E_1 - (a_0, a_{2k})$ and any edge in E_3 do not nest.

Secondly, since the length of (a_0, a_{2k}) is $2k$ and the length of the edge in E_2 is $2k + 1$, any two edges of second queue do not nest. $\square$

For example, $GP(5, 2)$ is a 2-queue graph. The vertex ordering $a_0, a_1, a_2, a_3, a_4, b_0, b_1, b_2, b_3, b_4$ permits a 2-queue layout. Figure 3 shows the result of two queues for placing the edges of $GP(5, 2)$, where edges above the vertices are assigned to the first queue and edges below the vertices are assigned to the second queue.

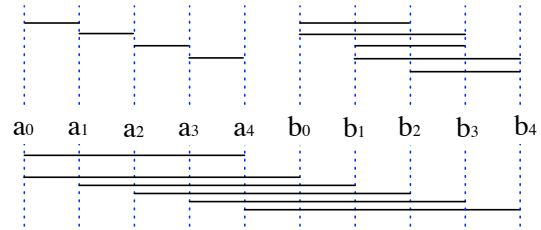


Figure 3: A 2-queue layout of $GP(5, 2)$.

The following two theorems are the basis to prove the low bound of $qn(GP(2k + 1, k))$.

Theorem 3 (Heath and Rosenberg [9])

A graph G is a 1-queue graph if and only if G is an arched leveled-planar graph.

Theorem 4 (Wagner, 1937) A graph G is planar if and only if neither the complete graph K_5 nor the complete bipartite graph $K_{3,3}$ is a minor of G .

Lemma 5 $GP(2k+1, k)$ is nonplanar while $k \geq 2$.

Proof. Assume that $k \geq 2$. First, we are to begin with a graph $GP(2k+1, k)$, contract all edges in E_2 to produce the graph $CI_{2k+1}(1, k)$. Then we obtain the graph K_5 after contracting $(a_0, a_1), (a_1, a_2), \dots, (a_{k-3}, a_{k-2})$ and $(a_k, a_{k+1}), (a_{k+1}, a_{k+2}), \dots, (a_{2k-3}, a_{2k-2})$.

Due to K_5 is a minor of $GP(2k+1, k)$, $GP(2k+1, k)$ is nonplanar by Theorem 4. $\square$

For example, we are to begin with a graph $GP(7, 3)$, contract $(a_0, b_0), (a_1, b_1), (a_2, b_2), \dots, (a_6, b_6)$ to produce the graph $CI_7(1, 3)$ (as shown in Figure 4(a)). Next, we contract (a_0, a_1) and (a_3, a_4) to obtain the graph K_5 (as shown in Figure 4(b)).

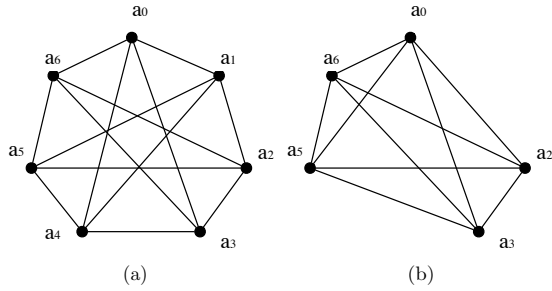


Figure 4: (a) $CI_7(1, 3)$; (b) K_5 .

Note that the arched leveled-planar graph is a subclass of the planar graph. By Lemma 5 and Theorem 3, we immediately obtain the following lemma.

Lemma 6 $qn(GP(2k+1, k)) \geq 2$ if $k \geq 2$.

Combining the results of Lemmas 2 and 6, we immediately obtain the queue number of $GP(2k+1, k)$ as the following theorem.

Theorem 7 $qn(GP(2k+1, k)) = 2$ if $k \geq 2$.

From the isomorphism of $GP(2k+1, k)$ and $GP(2k+1, 2)$, the following corollary directly follows from Theorem 7.

Corollary 8 $qn(GP(2k+1, 2)) = 2$ if $k \geq 2$.

3 Conclusion

As a concluding remark, we summarize Lemma 1, Theorem 7, and Corollary 8. The following are some results for the queue number of generalized Petersen graphs.

- (1) $qn(GP(3, 1)) = 1$.
- (2) $qn(GP(2k+1, k)) = 2$ if $k \geq 2$.
- (3) $qn(GP(2k+1, 2)) = 2$ if $k \geq 2$.

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